

Newman-Penrose Formalism Calculations

Stephani et al., Equation (17.14)

Metric

$$ds^2 = -\frac{e^{(-2t)}}{z}dt \otimes dt + \frac{\sqrt{3}(\sqrt{2}\sqrt{4z^2 - 4z + 3} + \sqrt{3})e^{(-2t)}}{3z}dt \otimes dy + 2e^{(2y)}dx \otimes dx + \frac{\sqrt{3}(\sqrt{2}\sqrt{4z^2 - 4z + 3} + \sqrt{3})e^{(-2t)}}{3z}dy \otimes dt \\ + \frac{\sqrt{3}(4\sqrt{3}z - 2\sqrt{2}\sqrt{4z^2 - 4z + 3} - 3\sqrt{3})e^{(-2t)}}{3z}dy \otimes dy + \left(\frac{2\sqrt{3}\sqrt{2}e^{(-2t)}}{3\sqrt{4z^2 - 4z + 3}z}\right)dy \otimes dz + \left(\frac{2\sqrt{3}\sqrt{2}e^{(-2t)}}{3\sqrt{4z^2 - 4z + 3}z}\right)dz \otimes dy + \left(\frac{2}{4z^4e^{(2t)} - 4z^3e^{(2t)} + 3z^2e^{(2t)}}\right)dz \otimes dz$$

Coordinates

$$(x^\mu) = (t, x, y, z)$$

Metric Functions

$$A = \frac{\sqrt{\frac{1}{2}\sqrt{4z^2 + 4z + 3}}}{2\sqrt{z}}$$

$$B = \frac{\sqrt{3}}{2\sqrt{z}}$$

$$H = \frac{2}{3}\sqrt{3}\sqrt{\frac{1}{2}\sqrt{4z^2 - 4z + 3}} + 1$$

Null Tetrad

Covariant components

$$l = \left(\frac{\sqrt{2}e^{(-t)}}{2\sqrt{z}}\right)dt + \left(-\frac{\sqrt{3}\sqrt{2}\sqrt{\frac{1}{2}\sqrt{4z^2 - 4z + 3}}e^{(-t)}}{3\sqrt{z}} + \frac{1}{3}\sqrt{3}\sqrt{2}e^{(-t)} - \frac{\sqrt{2}e^{(-t)}}{2\sqrt{z}}\right)dy + \left(\frac{\sqrt{2}\sqrt{\frac{1}{2}}e^{(-t)}}{\sqrt{4z^2 - 4z + 3}z}\right)dz$$

$$n = \left(\frac{\sqrt{2}e^{(-t)}}{2\sqrt{z}} \right) dt + \left(-\frac{\sqrt{3}\sqrt{2}\sqrt{\frac{1}{2}\sqrt{4z^2-4z+3}}e^{(-t)}}{3\sqrt{z}} - \frac{1}{3}\sqrt{3}\sqrt{2}e^{(-t)} - \frac{\sqrt{2}e^{(-t)}}{2\sqrt{z}} \right) dy + \left(-\frac{\sqrt{2}\sqrt{\frac{1}{2}}e^{(-t)}}{\sqrt{4z^2-4z+3}z} \right) dz$$

$$m = (e^y) dx + \left(2i\sqrt{\frac{1}{3}}\sqrt{z}e^{(-t)} \right) dy$$

$$\bar{m} = (e^y) dx + \left(-2i\sqrt{\frac{1}{3}}\sqrt{z}e^{(-t)} \right) dy$$

Contravariant components

$$l = \left(-\frac{1}{2}\sqrt{2}\sqrt{z}e^t \right) dt + \left(\frac{2z^3e^t}{\sqrt{4z^2-4z+3}} - \frac{2z^2e^t}{\sqrt{4z^2-4z+3}} + \frac{3ze^t}{2\sqrt{4z^2-4z+3}} \right) dz$$

$$n = \left(-\frac{1}{2}\sqrt{2}\sqrt{z}e^t \right) dt + \left(-\frac{2z^3e^t}{\sqrt{4z^2-4z+3}} + \frac{2z^2e^t}{\sqrt{4z^2-4z+3}} - \frac{3ze^t}{2\sqrt{4z^2-4z+3}} \right) dz$$

$$m = \left(\frac{i\sqrt{2}\sqrt{4z^2-4z+3}e^t}{4\sqrt{z}} + \frac{i\sqrt{3}e^t}{4\sqrt{z}} \right) dt + \left(\frac{1}{2}e^{(-y)} \right) dx + \left(\frac{i\sqrt{3}e^t}{4\sqrt{z}} \right) dy + \left(-\frac{i\sqrt{2}z^{\frac{5}{2}}e^t}{\sqrt{4z^2-4z+3}} + \frac{i\sqrt{2}z^{\frac{3}{2}}e^t}{\sqrt{4z^2-4z+3}} - \frac{3i\sqrt{2}\sqrt{z}e^t}{4\sqrt{4z^2-4z+3}} \right) dz$$

$$\bar{m} = \left(-\frac{i\sqrt{2}\sqrt{4z^2-4z+3}e^t}{4\sqrt{z}} - \frac{i\sqrt{3}e^t}{4\sqrt{z}} \right) dt + \left(\frac{1}{2}e^{(-y)} \right) dx + \left(-\frac{i\sqrt{3}e^t}{4\sqrt{z}} \right) dy + \left(\frac{i\sqrt{2}z^{\frac{5}{2}}e^t}{\sqrt{4z^2-4z+3}} - \frac{i\sqrt{2}z^{\frac{3}{2}}e^t}{\sqrt{4z^2-4z+3}} + \frac{3i\sqrt{2}\sqrt{z}e^t}{4\sqrt{4z^2-4z+3}} \right) dz$$

Directional Derivatives

$$D = -\frac{1}{2}\sqrt{2}\sqrt{z}e^t\partial_t + \frac{2z^3e^t}{\sqrt{4z^2-4z+3}} - \frac{2z^2e^t}{\sqrt{4z^2-4z+3}} + \frac{3ze^t}{2\sqrt{4z^2-4z+3}}\partial_z$$

$$\Delta = -\frac{1}{2}\sqrt{2}\sqrt{z}e^t\partial_t - \frac{2z^3e^t}{\sqrt{4z^2-4z+3}} + \frac{2z^2e^t}{\sqrt{4z^2-4z+3}} - \frac{3ze^t}{2\sqrt{4z^2-4z+3}}\partial_z$$

$$\delta = \frac{i\sqrt{2}\sqrt{4z^2-4z+3}e^t}{4\sqrt{z}} + \frac{i\sqrt{3}e^t}{4\sqrt{z}}\partial_t + \frac{1}{2}e^{(-y)}\partial_x + \frac{i\sqrt{3}e^t}{4\sqrt{z}}\partial_y - \frac{i\sqrt{2}z^{\frac{5}{2}}e^t}{\sqrt{4z^2-4z+3}} + \frac{i\sqrt{2}z^{\frac{3}{2}}e^t}{\sqrt{4z^2-4z+3}} - \frac{3i\sqrt{2}\sqrt{z}e^t}{4\sqrt{4z^2-4z+3}}\partial_z$$

$$\bar{\delta} = -\frac{i\sqrt{2}\sqrt{4z^2-4z+3}e^t}{4\sqrt{z}} - \frac{i\sqrt{3}e^t}{4\sqrt{z}}\partial_t + \frac{1}{2}e^{(-y)}\partial_x - \frac{i\sqrt{3}e^t}{4\sqrt{z}}\partial_y + \frac{i\sqrt{2}z^{\frac{5}{2}}e^t}{\sqrt{4z^2-4z+3}} - \frac{i\sqrt{2}z^{\frac{3}{2}}e^t}{\sqrt{4z^2-4z+3}} + \frac{3i\sqrt{2}\sqrt{z}e^t}{4\sqrt{4z^2-4z+3}}\partial_z$$

Spin Coefficients

$$\rho = -\frac{z^2e^t}{2\sqrt{4z^2-4z+3}} - \frac{1}{4}\sqrt{2}\sqrt{z}e^t + \frac{ze^t}{2\sqrt{4z^2-4z+3}} - \frac{3e^t}{8\sqrt{4z^2-4z+3}}$$

$$\sigma = \frac{z^2e^t}{2\sqrt{4z^2-4z+3}} + \frac{1}{4}\sqrt{2}\sqrt{z}e^t - \frac{ze^t}{2\sqrt{4z^2-4z+3}} + \frac{3e^t}{8\sqrt{4z^2-4z+3}}$$

$$\kappa = -\frac{i\sqrt{2}z^{\frac{3}{2}}e^t}{4\sqrt{4z^2-4z+3}} + \frac{1}{2}ie^t + \frac{i\sqrt{2}\sqrt{z}e^t}{4\sqrt{4z^2-4z+3}} - \frac{3i\sqrt{2}e^t}{16\sqrt{4z^2-4z+3}\sqrt{z}} - \frac{1}{4}ie^t$$

$$\mu = -\frac{z^2e^t}{2\sqrt{4z^2-4z+3}} + \frac{1}{4}\sqrt{2}\sqrt{z}e^t + \frac{ze^t}{2\sqrt{4z^2-4z+3}} - \frac{3e^t}{8\sqrt{4z^2-4z+3}}$$

$$\lambda = \frac{z^2e^t}{2\sqrt{4z^2-4z+3}} - \frac{1}{4}\sqrt{2}\sqrt{z}e^t - \frac{ze^t}{2\sqrt{4z^2-4z+3}} + \frac{3e^t}{8\sqrt{4z^2-4z+3}}$$

$$\nu = -\frac{i\sqrt{2}z^{\frac{3}{2}}e^t}{4\sqrt{4z^2-4z+3}} - \frac{1}{2}ie^t + \frac{i\sqrt{2}\sqrt{z}e^t}{4\sqrt{4z^2-4z+3}} - \frac{3i\sqrt{2}e^t}{16\sqrt{4z^2-4z+3}\sqrt{z}} + \frac{1}{4}ie^t$$

$$\alpha = -\frac{1}{4}ie^t + \frac{i\sqrt{3}e^t}{8\sqrt{z}} + \frac{1}{8}ie^t$$

$$\beta = \frac{1}{4} i z e^t + \frac{i \sqrt{3} e^t}{8 \sqrt{z}} - \frac{1}{8} i e^t$$

$$\tau = \frac{3i \sqrt{2} z^{\frac{3}{2}} e^t}{4 \sqrt{4 z^2 - 4 z + 3}} - \frac{3i \sqrt{2} \sqrt{z} e^t}{4 \sqrt{4 z^2 - 4 z + 3}} + \frac{i \sqrt{3} e^t}{4 \sqrt{z}} + \frac{9i \sqrt{2} e^t}{16 \sqrt{4 z^2 - 4 z + 3} \sqrt{z}}$$

$$\pi = \frac{3i \sqrt{2} z^{\frac{3}{2}} e^t}{4 \sqrt{4 z^2 - 4 z + 3}} - \frac{3i \sqrt{2} \sqrt{z} e^t}{4 \sqrt{4 z^2 - 4 z + 3}} + \frac{i \sqrt{3} e^t}{4 \sqrt{z}} + \frac{9i \sqrt{2} e^t}{16 \sqrt{4 z^2 - 4 z + 3} \sqrt{z}}$$

$$\epsilon = -\frac{z^2 e^t}{2 \sqrt{4 z^2 - 4 z + 3}} + \frac{1}{4} \sqrt{2} \sqrt{z} e^t + \frac{z e^t}{2 \sqrt{4 z^2 - 4 z + 3}} - \frac{3 e^t}{8 \sqrt{4 z^2 - 4 z + 3}}$$

$$\gamma = -\frac{z^2 e^t}{2 \sqrt{4 z^2 - 4 z + 3}} - \frac{1}{4} \sqrt{2} \sqrt{z} e^t + \frac{z e^t}{2 \sqrt{4 z^2 - 4 z + 3}} - \frac{3 e^t}{8 \sqrt{4 z^2 - 4 z + 3}}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = 0$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = 0$$

$$\Lambda = 0$$

Weyl Tensor Components

$$\Psi_0 = \frac{\sqrt{3}\sqrt{2}ze^{(2t)}}{8\sqrt{4z^2-4z+3}} - \frac{1}{4}\sqrt{3}\sqrt{z}e^{(2t)} - \frac{\sqrt{3}\sqrt{2}e^{(2t)}}{8\sqrt{4z^2-4z+3}} + \frac{\sqrt{3}e^{(2t)}}{8\sqrt{z}} + \frac{3\sqrt{3}\sqrt{2}e^{(2t)}}{32\sqrt{4z^2-4z+3}z}$$

$$\Psi_1 = -\frac{i\sqrt{3}z^{\frac{3}{2}}e^{(2t)}}{4\sqrt{4z^2-4z+3}} - \frac{1}{8}i\sqrt{3}\sqrt{2}e^{(2t)} + \frac{i\sqrt{3}\sqrt{z}e^{(2t)}}{4\sqrt{4z^2-4z+3}} - \frac{3i\sqrt{3}e^{(2t)}}{16\sqrt{4z^2-4z+3}\sqrt{z}}$$

$$\Psi_2 = \frac{3\sqrt{3}\sqrt{2}\sqrt{4z^2-4z+3}e^{(2t)}}{32z} + \frac{3e^{(2t)}}{8z}$$

$$\Psi_3 = -\frac{i\sqrt{3}z^{\frac{3}{2}}e^{(2t)}}{4\sqrt{4z^2-4z+3}} + \frac{1}{8}i\sqrt{3}\sqrt{2}e^{(2t)} + \frac{i\sqrt{3}\sqrt{z}e^{(2t)}}{4\sqrt{4z^2-4z+3}} - \frac{3i\sqrt{3}e^{(2t)}}{16\sqrt{4z^2-4z+3}\sqrt{z}}$$

$$\Psi_4 = \frac{\sqrt{3}\sqrt{2}ze^{(2t)}}{8\sqrt{4z^2-4z+3}} + \frac{1}{4}\sqrt{3}\sqrt{z}e^{(2t)} - \frac{\sqrt{3}\sqrt{2}e^{(2t)}}{8\sqrt{4z^2-4z+3}} - \frac{\sqrt{3}e^{(2t)}}{8\sqrt{z}} + \frac{3\sqrt{3}\sqrt{2}e^{(2t)}}{32\sqrt{4z^2-4z+3}z}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"